

Netflix Recommendation System using Collaborative Filtering and Matrix Factorization

1. Problem Understanding

Recommendation systems play a critical role in modern digital platforms by helping users discover relevant content from large catalogs. The original Netflix Prize Dataset contains 100,480,507 ratings from 480,189 users across 17,770 movies, collected between 1998 and 2005.

The dataset is highly sparse (~98.8% sparsity), making it a challenging benchmark for large-scale recommendation systems providing a large-scale recommendation problem involving millions of user-movie interactions and serves as a benchmark dataset for recommender system research.

The objective of this project was to develop a personalized movie recommendation system capable of predicting unseen ratings and generating relevant movie recommendations. Multiple recommendation approaches were implemented and compared under different evaluation settings to identify the most effective model.

The project focused on:

- Rating prediction accuracy
- Recommendation ranking quality
- Temporal evaluation strategies
- Cold-start analysis
- Personalized recommendation generation

Models evaluated:

- Item-Based Collaborative Filtering
- Singular Value Decomposition (SVD)
- SVD++

2. Exploratory Data Analysis

Dataset Statistics

Metric	Value
Ratings	2,000,000
Users	358,375
Movies	17,324
Original Movies	17,770

The original Netflix Prize Dataset contains more than 100 million ratings. Due to computational limitations, a representative sample of 2 million ratings was created for experimentation.

Despite the reduction in ratings, the sample retained 17,324 of the original 17,770 movies, corresponding to approximately 97.5% movie coverage. This ensured that the sample remained representative of the overall catalog while significantly reducing computational cost.

Rating Distribution

The average rating in the dataset was 3.61, with a median rating of 4. Most ratings were concentrated between 3 and 5 stars, indicating that users generally rate content they enjoy.

User Activity Analysis

User activity followed a long-tail distribution. Most users contributed only a small number of ratings, while a small subset of highly active users generated a large proportion of interactions.

Movie Popularity Analysis

Movie popularity was highly imbalanced. A small number of movies received most ratings, while the majority of movies received relatively few interactions.

Key Findings

- Ratings are positively skewed.
 - User activity is highly uneven.
 - Movie popularity follows a long-tail distribution.
 - The user-item matrix is extremely sparse.
 - Temporal information enables realistic evaluation.
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3. Methodology

Data Preparation

The Netflix data files were processed and converted into a standard user-item-rating format.

A modeling dataset containing 2 million ratings was created for experimentation.

Train-Test Split Strategies

Three evaluation strategies were explored.

Random Split

- 80% Training Data
- 20% Testing Data

This provides a useful baseline but may introduce temporal leakage.

Global Temporal Split

- Earlier ratings → Training
- Later ratings → Testing

This better reflects real-world deployment conditions.

Per-User Temporal Split

For each user:

- Older ratings - Training
- Recent ratings - Testing

This strategy most closely simulates practical recommendation systems and was selected as the primary evaluation approach.

Split Diagnostics

Analysis of the temporal split revealed:

- 57,336 users appeared only in the testing period.
- Approximately 48% of testing ratings belonged to cold users.

These observations highlight the challenges of real-world recommendation systems and motivated the use of per-user temporal evaluation.

4. Model Design

Item-Based Collaborative Filtering

Item-Based Collaborative Filtering predicts ratings using similarities between movies. It serves as a strong collaborative filtering baseline but is sensitive to data sparsity.

Singular Value Decomposition (SVD)

SVD learns latent user and movie representations from historical interactions. It is particularly effective for sparse recommendation datasets and can capture hidden preference patterns.

SVD++

SVD++ extends traditional SVD by incorporating implicit interaction information in addition to explicit ratings.

Hyperparameter Tuning

The best SVD configuration was obtained using:

Parameter	Value
n_factors	100
n_epochs	50
lr_all	0.003
reg_all	0.05

5. Evaluation Metrics

Recommendation performance was evaluated using both prediction and ranking metrics.

RMSE

Root Mean Squared Error (RMSE) was used to evaluate rating prediction accuracy.

Lower RMSE indicates better performance.

MAP@10

Mean Average Precision at 10 (MAP@10) was used to evaluate recommendation ranking quality.

A movie was considered relevant if:

Actual Rating ≥ 4

Additional Metrics

- Precision@10
- Recall@10

These metrics provide additional insight into recommendation effectiveness.

6. Experimental Results

RMSE Comparison

Model	RMSE
Item-CF Random	1.088
SVD Random	0.985
Item-CF Global Temporal	1.125
SVD Global Temporal	1.034
Item-CF Per User Temporal	1.154

SVD Per User Temporal	0.973
SVD++	0.979
Tuned SVD	0.970

Findings

- Matrix factorization methods consistently outperformed Item-Based Collaborative Filtering.
- Temporal evaluation produced higher prediction errors than random evaluation.
- Per-user temporal evaluation provided a more realistic estimate of recommendation performance.
- Hyperparameter tuning improved SVD performance.
- Tuned SVD achieved the best overall RMSE.

Ranking Evaluation

Initial ranking evaluation produced the following results:

Metric	Score
Precision@10	0.594
Recall@10	0.999
MAP@10	0.644

However, many users contained very small test histories, causing inflated ranking metrics.

To obtain a more realistic estimate, evaluation was repeated using only users with at least 10 test interactions.

Metric	Score
Precision@10	0.437
Recall@10	0.898
MAP@10	0.597

These refined results provide a more reliable measure of recommendation quality.

Cold Start Analysis

Segment	Ratings	RMSE
Cold	116,863	1.697
Light	124,001	1.737
Medium	26,493	1.910
Heavy	446	3.211

Cold-start users remain a significant challenge because limited interaction history restricts the model's ability to accurately learn user preferences.

7. Recommendation Examples

Top recommendations generated using the tuned SVD model included:

Rank	Movie	Predicted Rating
1	The West Wing: Season 3	4.748

2	Lord of the Rings: The Return of the King	4.700
3	The Lord of the Rings: The Fellowship of the Ring	4.698
4	Spirited Away	4.695
5	CSI: Season 4	4.684
6	The Simpsons: Season 5	4.673
7	The Sopranos: Season 3	4.670
8	The Work of Director Michel Gondry	4.666
9	Lord of the Rings: The Return of the King (Special Edition)	4.656
10	Lost: Season 1	4.636

The recommendations span multiple genres and formats, indicating that the model successfully learned meaningful user preference patterns rather than relying solely on popularity.

8. Key Insights

Insight 1

A sample containing only 2 million ratings preserved approximately 97.5% of the movie catalog, demonstrating that representative sampling can significantly reduce computational cost while maintaining coverage.

Insight 2

Matrix factorization methods consistently outperformed neighborhood-based collaborative filtering approaches.

Insight 3

Temporal evaluation produced more realistic performance estimates than random train-test splitting.

Insight 4

Approximately 48% of temporal testing ratings originated from cold users, highlighting the practical importance of cold-start challenges.

Insight 5

Recommendation ranking metrics can become inflated when users contain very few testing interactions. Filtering users with sufficient test history produced more reliable evaluation results.

9. Future Improvements

Potential future directions include:

- Alternating Least Squares (ALS)
- Neural Collaborative Filtering
- Hybrid Recommendation Systems
- Content-Based Features
- Diversity-Aware Recommendation Objectives
- Explainable Recommendation Systems
- Training on larger portions of the Netflix dataset

10. Conclusion

This project developed and evaluated multiple recommendation approaches using the Netflix Prize Dataset.

Item-Based Collaborative Filtering, SVD, and SVD++ were compared under random, temporal, and per-user temporal evaluation strategies.

The results demonstrated that matrix factorization approaches consistently outperform neighborhood-based methods. The tuned SVD model achieved the best predictive performance with an RMSE of 0.9703 while also producing high-quality Top-10 recommendations.

The project further highlighted the importance of realistic temporal evaluation, careful ranking metric design, and cold-start analysis. Overall, the study demonstrates the effectiveness of latent factor models for large-scale recommendation systems and provides a strong foundation for future recommender system development.